

Week 3 - Day 2: Linear Regression & Model Evaluation

5-Minute Review Notes

Day Goal: Understand how Linear Regression learns a line, how to interpret slope and intercept, and how to evaluate regression predictions using MAE, MSE, RMSE, and R-Squared.

1. Linear Regression Recap

Linear Regression models the relationship between one or more input features and a continuous numerical target. With one feature, it learns a straight line that best represents the relationship in the training data.

Core equation: $y = mx + b$

y = predicted target, x = feature, m = slope/coefficient, b = intercept.

For the Day 2 salary model:

X: Experience_Years

y: Salary

2. Slope / Coefficient

The slope tells how much the predicted target changes when the feature increases by one unit.

```
model.coef_[0] -> 3747.85
```

Interpretation: according to the fitted model, each additional year of experience is associated with approximately **\$3,747.85 higher predicted salary**.

3. Intercept

The intercept is the predicted target value when the feature equals zero.

```
model.intercept_ -> 36226.36
```

Interpretation: mathematically, the model predicts about **\$36,226.36** when Experience_Years = 0. This may not be practically meaningful because the dataset contains only 1-20 years of experience and no zero-year example.

4. Actual, Predicted, and Residual

A residual/error measures the difference between an actual value and the model prediction.

Error = Actual - Predicted

Example: Actual = \$80,000 and Predicted = \$76,000 gives Error = +\$4,000 and Absolute Error = \$4,000. The positive sign means the model underpredicted.

5. MAE - Mean Absolute Error

MAE is the average absolute prediction error. It is easy to interpret because it stays in the original target units.

Day 2 result: MAE = 1045.85

Interpretation: the model's salary predictions are off by about \$1,045.85 on average on the test data.

6. MSE - Mean Squared Error

MSE squares each residual before averaging. Squaring causes larger errors to receive a stronger penalty.

```
Day 2 result: MSE = 1,561,674.37
```

MSE is not directly interpretable as dollars because its units are squared salary units.

7. RMSE - Root Mean Squared Error

RMSE is the square root of MSE. It returns the metric to the original target units while still being sensitive to larger errors.

Day 2 result: RMSE = 1249.67

Interpretation: the model has a typical error magnitude of roughly \$1,249.67, with larger mistakes weighted more strongly than under MAE.

8. R-Squared

R-Squared measures performance relative to a baseline that always predicts the mean target value.

```
R2 = 1 -> perfect fit
R2 = 0 -> no better than predicting the mean
R2 < 0 -> worse than the mean baseline
```

Day 2 result: R-Squared = 0.9980. This indicates an excellent fit on this test set and that the model explains about 99.8% of the variation in salary relative to the mean baseline. It does not mean the model is 99.8% accurate.

9. Model Evaluation Summary

Metric	Day 2 Result	Interpretation	Better
MAE	\$1,045.85	Average absolute error	Lower
MSE	1,561,674.37	Average squared error	Lower
RMSE	\$1,249.67	Error magnitude in original units	Lower
R-Squared	0.9980	Fit relative to mean baseline	Higher

10. Day 2 Code Workflow

```
X = df[["Experience_Years"]]
y = df["Salary"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

Evaluation:

```
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = root_mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)
```

11. Day 2 Prediction Results

Experience Years	Actual Salary	Predicted Salary
1	\$42,000	\$39,974.21
18	\$103,000	\$103,687.68
16	\$96,000	\$96,191.98
2	\$45,000	\$43,722.06

12. Single Prediction - 10.5 Years Experience

A one-row DataFrame was used so the input remained 2D and preserved the feature name:

```
single_input = pd.DataFrame({"Experience_Years": [10.5]})
single_pred = model.predict(single_input)
```

Prediction: \$75,578.80

This is reasonable because the dataset shows approximately \$73,000 at 10 years and \$77,000 at 11 years. Since 10.5 lies inside the observed range, this is an example of **interpolation**.

13. Interpolation vs Extrapolation

Interpolation: prediction inside the range of observed feature values. Example: 10.5 years when training data covers 1-20 years.

Extrapolation: prediction outside the observed range. Example: asking a model trained on 1-20 years to predict salary for 50 years of experience. Extrapolated predictions can be less reliable.

14. Overfitting - First Introduction

If training performance is excellent but testing performance is much worse, the model may have learned the training data too specifically and may not generalize well.

Example:

Training R-Squared = 0.98

Testing R-Squared = 0.60

This gap is a warning sign of possible overfitting.

15. Day 2 Assessment - Correct Answers

Q1. What does the slope mean?

Each additional year of experience is associated with about \$3,747.85 higher predicted salary according to this model.

Q2. What does the intercept mean?

It is the predicted salary when Experience_Years = 0. It may not be practically meaningful because zero years is outside the observed training examples.

Q3. Actual \$80,000 vs predicted \$76,000?

Prediction error = +\$4,000 using Actual - Predicted. Absolute error = \$4,000.

Q4. What does MAE tell us?

The model's predictions are off by about \$1,045.85 on average.

Q5. Why does MSE punish large errors more?

Because residuals are squared, so larger errors grow much faster.

Q6. Why is RMSE easier to interpret than MSE?

RMSE returns the metric to the original target units.

Q7. Does R-Squared = 0.95 mean 95% accuracy?

No. R-Squared is not classification accuracy. It describes fit relative to a mean baseline and explained variation.

Q8. Which is better: MAE \$2,500 or \$7,500?

The \$2,500 model is better by MAE because its predictions are closer to actual values on average.

Q9. Why suspect overfitting with train R2 0.98 and test R2 0.60?

The model performs much better on training data than unseen testing data, suggesting weak generalization.

16. Quick Review Cheat Sheet

Concept	Remember
Linear Regression	Predicts a continuous numerical target using a learned linear relationship
Slope / coef_	Change in predicted y for a 1-unit increase in x
Intercept	Predicted y when x = 0
Residual	Actual - Predicted
MAE	Average absolute error; lower is better
MSE	Average squared error; strongly penalizes large errors
RMSE	Square root of MSE; original target units

Concept	Remember
R-Squared	Fit relative to mean baseline; higher is generally better
Interpolation	Prediction inside observed feature range
Extrapolation	Prediction outside observed feature range
Overfitting	Very strong training performance but much weaker unseen-test performance

17. Day 2 Completion Summary

Week 3 - Day 2: COMPLETE

Completed concepts:

- Linear Regression equation
- Slope / coefficient interpretation
- Intercept interpretation
- Actual vs predicted values
- Residual / prediction error
- MAE, MSE, RMSE, and R-Squared
- Metric interpretation
- Interpolation vs extrapolation
- First introduction to overfitting
- Single-value salary prediction
- End-to-end regression evaluation workflow

Progress: Week 1 Python - 7/7 complete | Week 2 Data Analysis - 7/7 complete | Week 3 Machine Learning - 2/7 complete